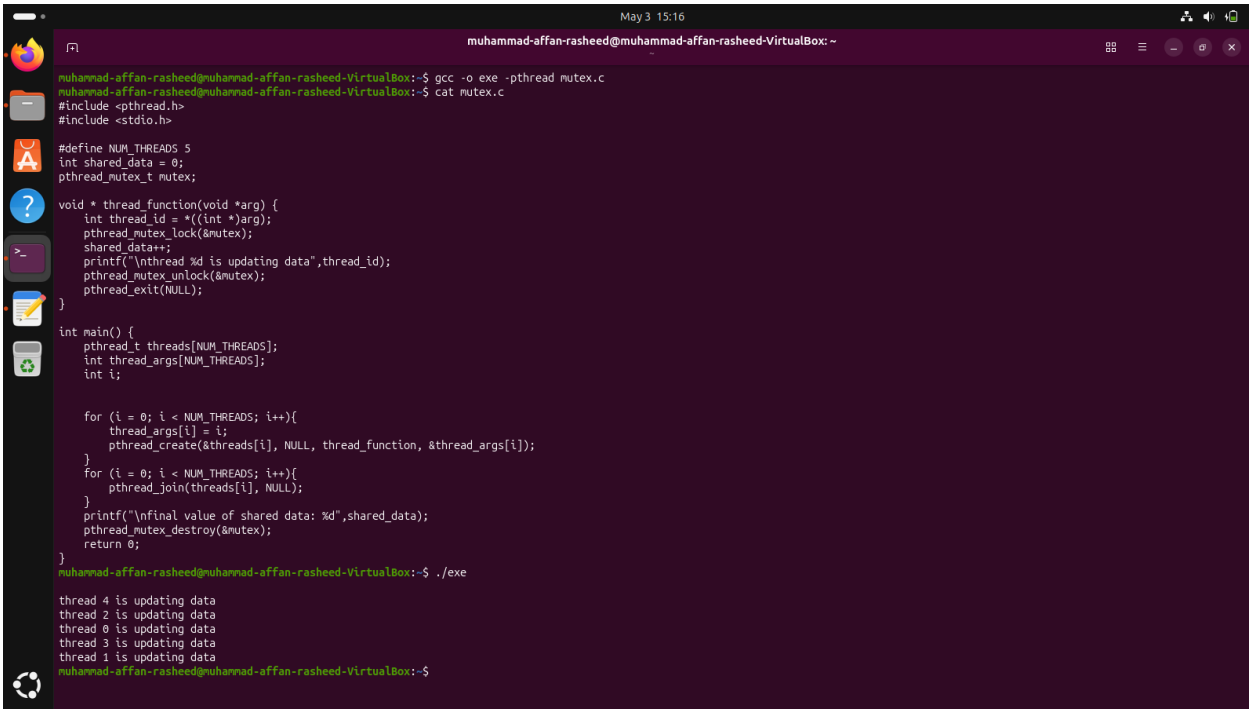
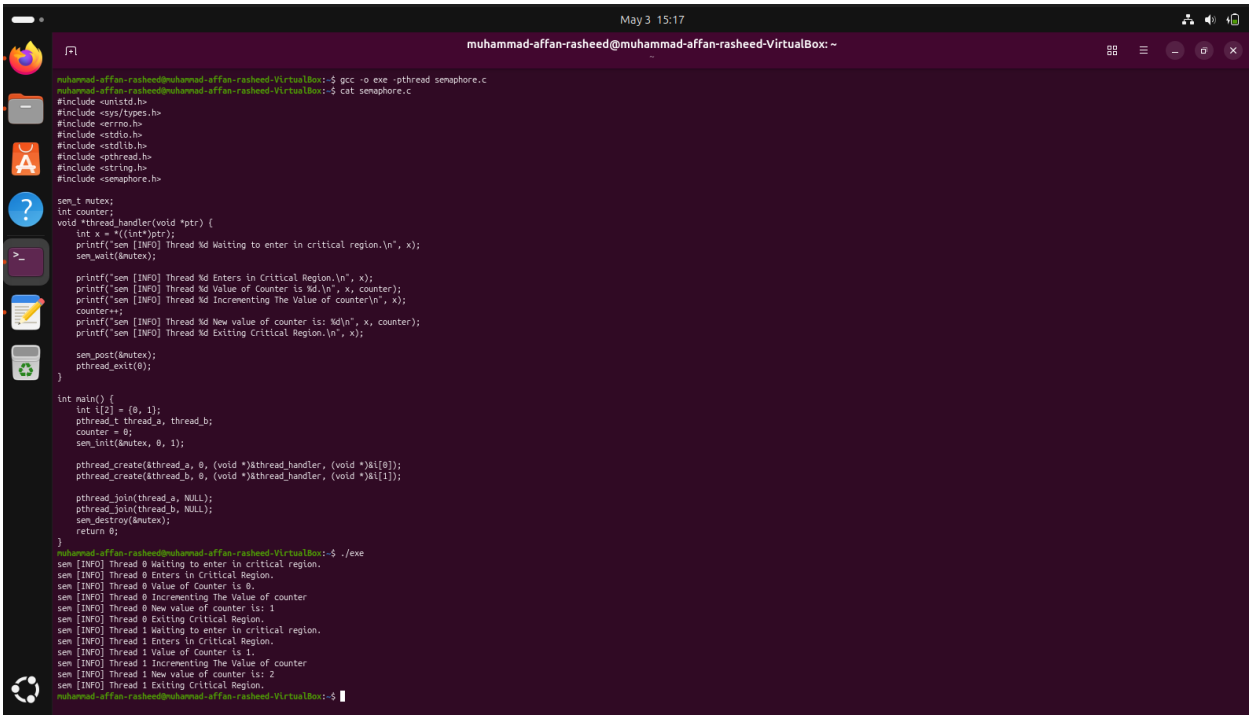
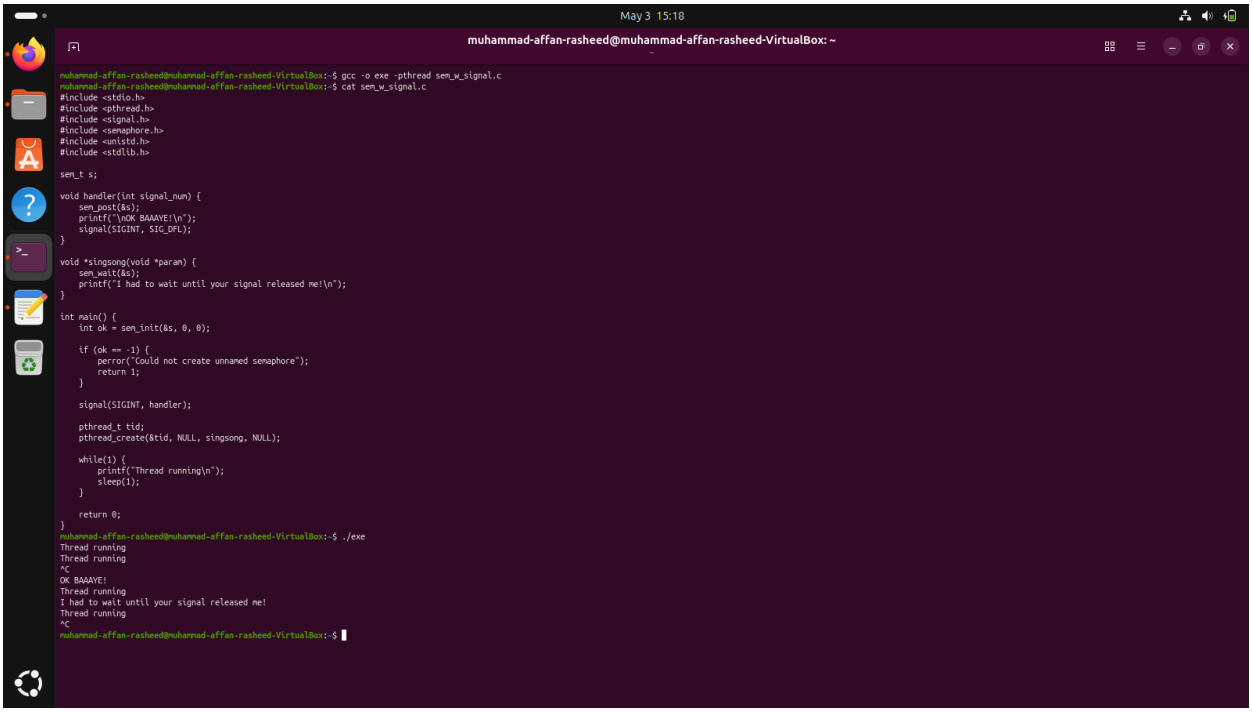
Simple mutex:



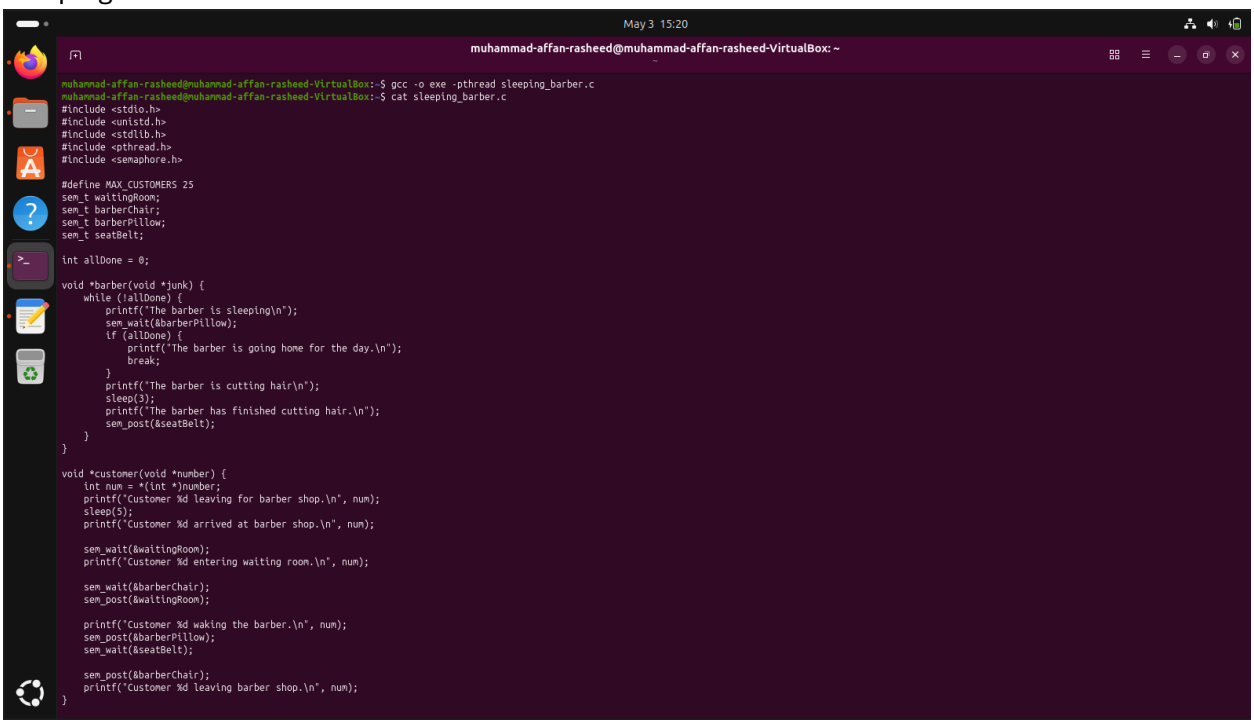
Semaphores:

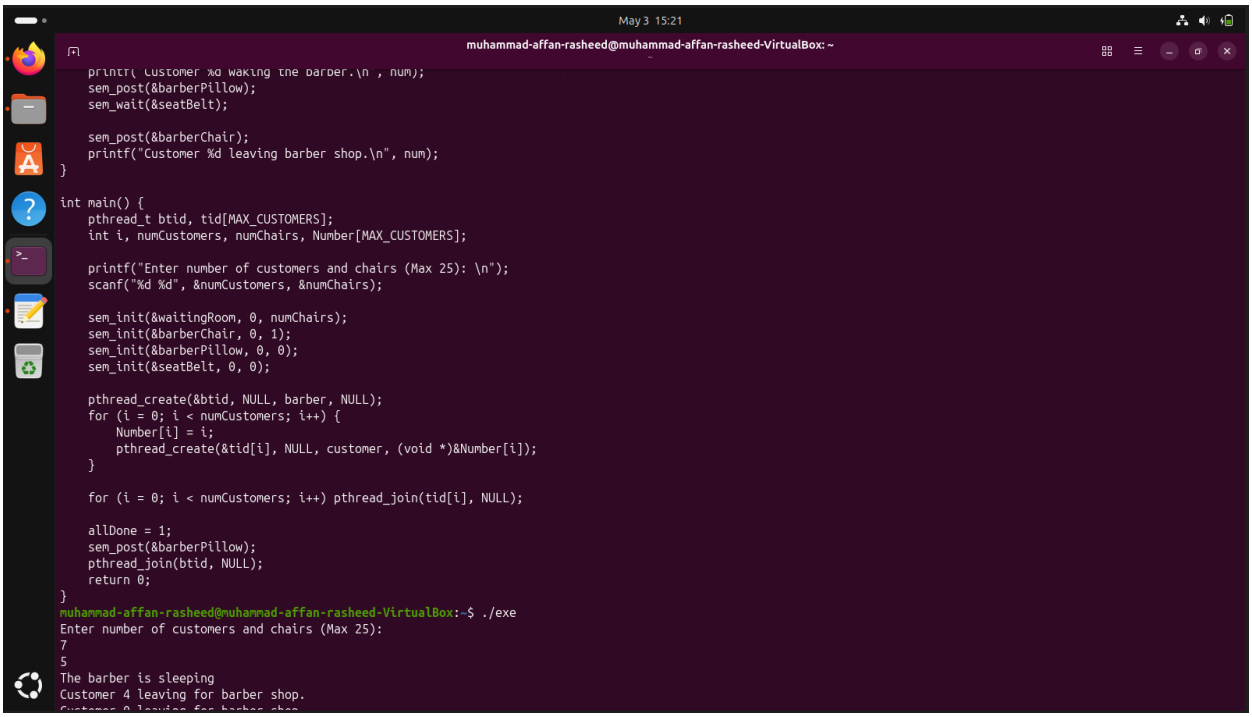


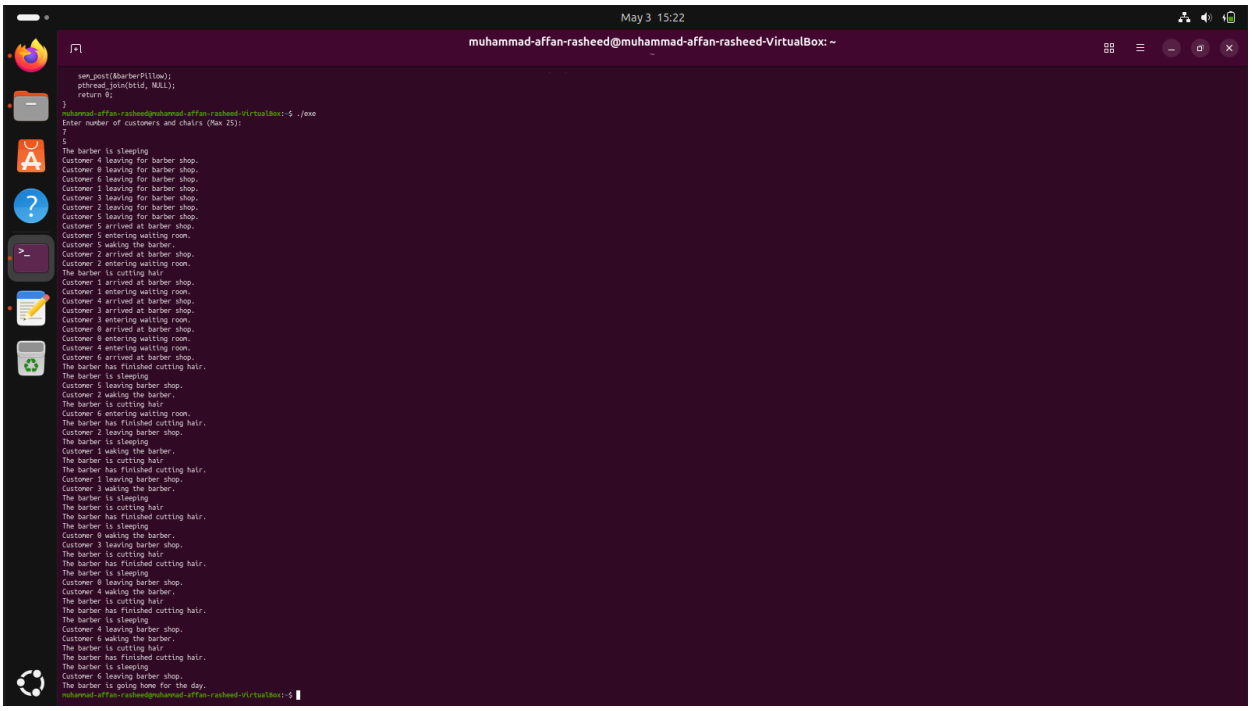
Semaphores with signals:



Sleeping barber:




May 3 15:21
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~
printf("Customer %d waking the barber.\n", num);
sem_post(&barberPillow);
sem_wait(&seatBelt);
sem_post(&barberChair);
printf("Customer %d leaving barber shop.\n", num);
}
int main() {
pthread_t btid, tid[MAX_CUSTOMERS];
int i, numCustomers, numChairs, Number[MAX_CUSTOMERS];
printf("Enter number of customers and chairs (Max 25): \n");
scanf("%d %d", &numCustomers, &numChairs);
sem_init(&waitingRoom, 0, numChairs);
sem_init(&barberChair, 0, 1);
sem_init(&barberPillow, 0, 0);
sem_init(&seatBelt, 0, 0);
pthread_create(&btid, NULL, barber, NULL);
for (i = 0; i < numCustomers; i++) {
Number[i] = i;
pthread_create(&tid[i], NULL, customer, (void *)&Number[i]);
}
for (i = 0; i < numCustomers; i++) pthread_join(tid[i], NULL);
allDone = 1;
sem_post(&barberPillow);
pthread_join(btid, NULL);
return 0;
}
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~$ ./exe
Enter number of customers and chairs (Max 25):
7
5
The barber is sleeping
Customer 4 leaving for barber shop.


May 3 15:22
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox: ~
sem_post(&barberPillow);
pthread_join(btid, NULL);
return 0;
}
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~$ ./exe
Enter number of customers and chairs (Max 25):
7
5
The barber is sleeping
Customer 4 leaving for barber shop.
Customer 0 leaving for barber shop.
Customer 6 leaving for barber shop.
Customer 1 leaving for barber shop.
Customer 3 leaving for barber shop.
Customer 2 leaving for barber shop.
Customer 5 leaving for barber shop.
Customer 5 arrived at barber shop.
Customer 5 entering waiting room.
Customer 5 waking the barber.
Customer 2 arrived at barber shop.
Customer 2 entering waiting room.
The barber is cutting hair
Customer 1 arrived at barber shop.
Customer 1 entering waiting room.
Customer 4 arrived at barber shop.
Customer 3 arrived at barber shop.
Customer 3 entering waiting room.
Customer 0 arrived at barber shop.
Customer 0 entering waiting room.
Customer 4 entering waiting room.
Customer 6 arrived at barber shop.
The barber has finished cutting hair.
The barber is sleeping
Customer 5 leaving barber shop.
Customer 2 waking the barber.
The barber is cutting hair
Customer 6 entering waiting room.
The barber has finished cutting hair.
Customer 2 leaving barber shop.
The barber is sleeping
Customer 1 waking the barber.
The barber is cutting hair
The barber has finished cutting hair.
Customer 1 leaving barber shop.
Customer 3 waking the barber.
The barber is sleeping
The barber is cutting hair
The barber has finished cutting hair.
The barber is sleeping
Customer 0 waking the barber.
Customer 3 leaving barber shop.
The barber is cutting hair
The barber has finished cutting hair.
The barber is sleeping
Customer 0 leaving barber shop.
Customer 4 waking the barber.
The barber is cutting hair
The barber has finished cutting hair.
The barber is sleeping
Customer 4 leaving barber shop.
Customer 6 waking the barber.
The barber is cutting hair
The barber has finished cutting hair.
The barber is sleeping
Customer 6 leaving barber shop.
The barber is going home for the day.
muhammad-affan-rasheed@muhammad-affan-rasheed-VirtualBox:~$